

B1 Fluids — Model Solutions, Trinity Term 2023

Question 1 — Bernoulli, hydrostatics and the sluice gate

(a)

Steady Euler, uniform ρ , gravity $-g\hat{\mathbf{z}}$:

$$(\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{1}{\rho}\nabla p - g\hat{\mathbf{z}}.$$

Apply the convective identity:

$$(\mathbf{v} \cdot \nabla)\mathbf{v} = \nabla\left(\frac{1}{2}\mathbf{v} \cdot \mathbf{v}\right) - \mathbf{v} \times (\nabla \times \mathbf{v}).$$

Write $g\hat{\mathbf{z}} = \nabla(gz)$:

$$\nabla\left(\frac{\mathbf{v} \cdot \mathbf{v}}{2} + \frac{p}{\rho} + gz\right) = \mathbf{v} \times (\nabla \times \mathbf{v}).$$

Dot with $\mathbf{v}$ (RHS $\perp \mathbf{v}$):

$$\mathbf{v} \cdot \nabla\left(\frac{\mathbf{v} \cdot \mathbf{v}}{2} + \frac{p}{\rho} + gz\right) = 0.$$

Directional derivative along a streamline vanishes:

$$\frac{\mathbf{v} \cdot \mathbf{v}}{2} + \frac{p}{\rho} + gz = \text{const. on a streamline.}$$

Interpretation. Energy per unit mass (kinetic + flow-work p/ρ + gravitational PE) is conserved along a streamline.

(b)

Gate closed, $\mathbf{v} = 0$: Euler reduces to hydrostatic balance:

$$\frac{1}{\rho} \frac{dp}{dz} = -g.$$

Integrate with $p(H) = p_{\text{atm}}$:

$$p(z) = p_{\text{atm}} + \rho g(H - z).$$

Net horizontal force per width is $\int_0^H (p - p_{\text{atm}}) dz$:

$$F_{\text{gate}} = L \int_0^H \rho g(H - z) dz.$$

Substitute $u = H - z$:

$$F_{\text{gate}} = L \rho g \int_0^H u du.$$

Evaluate:

$$F_{\text{closed}} = \frac{\rho g H^2 L}{2}.$$

(c)

Bernoulli on a surface streamline, upstream ($z = H, U, p_{\text{atm}}$) to downstream ($z = h, u, p_{\text{atm}}$):

$$\frac{U^2}{2} + gH = \frac{u^2}{2} + gh.$$

Mass conservation (equal width L , uniform profiles):

$$UH = uh.$$

Eliminate $u = UH/h$:

$$\frac{U^2}{2} + gH = \frac{U^2 H^2}{2h^2} + gh.$$

Divide by gH , set $\hat{h} = h/H$, $F^2 = U^2/(gH)$:

$$\frac{F^2}{2} + 1 = \frac{F^2}{2\hat{h}^2} + \hat{h}.$$

Multiply by $2\hat{h}^2$:

$$F^2 \hat{h}^2 + 2\hat{h}^2 = F^2 + 2\hat{h}^3.$$

Rearrange as cubic in $\hat{h}$:

$$2\hat{h}^3 - (F^2 + 2)\hat{h}^2 + F^2 = 0.$$

Test trivial root $\hat{h} = 1$: $2 - (F^2 + 2) + F^2 = 0$. Factor $(\hat{h} - 1)$:

$$2\hat{h}^3 - (F^2 + 2)\hat{h}^2 + F^2 = (\hat{h} - 1)(2\hat{h}^2 - F^2\hat{h} - F^2).$$

Non-trivial roots solve:

$$2\hat{h}^2 - F^2\hat{h} - F^2 = 0.$$

Apply the quadratic formula:

$$\hat{h} = \frac{F^2 \pm \sqrt{F^4 + 8F^2}}{4}.$$

Factor F^2 from the surd, $F > 0$:

$$\hat{h} = \frac{F^2 \pm F\sqrt{F^2 + 8}}{4}.$$

Reject the negative root (gives $\hat{h} < 0$):

$$\boxed{\hat{h} = 1 \quad \text{or} \quad \hat{h} = \frac{F^2 + F\sqrt{F^2 + 8}}{4}}.$$

(d)

Steady horizontal momentum budget over CV between upstream ($x = -X$) and downstream ($x = +X$) sections:

$$\dot{M}_{\text{out}} - \dot{M}_{\text{in}} = \sum F_x^{\text{ext}}.$$

Momentum fluxes per unit width L , integrated over depth:

$$\dot{M}_{\text{in}} = \rho U^2 H L, \quad \dot{M}_{\text{out}} = \rho u^2 h L.$$

Use $uh = UH$, hence $u^2h = U^2H^2/h$:

$$\dot{M}_{\text{out}} - \dot{M}_{\text{in}} = \rho U^2 H L \left(\frac{H}{h} - 1 \right).$$

Express in terms of $\hat{h}$:

$$\dot{M}_{\text{out}} - \dot{M}_{\text{in}} = \frac{\rho U^2 H L (1 - \hat{h})}{\hat{h}}.$$

End-section pressures are hydrostatic (parallel streamlines); collect external forces:

$$\sum F_x^{\text{ext}} = \frac{\rho g L}{2} (H^2 - h^2) - F_{\text{gate}}.$$

Equate momentum budget:

$$\frac{\rho U^2 H L (1 - \hat{h})}{\hat{h}} = \frac{\rho g L}{2} (H^2 - h^2) - F_{\text{gate}}.$$

Solve for F_{gate} :

$$F_{\text{gate}} = \frac{\rho g L}{2} (H^2 - h^2) - \frac{\rho U^2 H L (1 - \hat{h})}{\hat{h}}.$$

Factor $H^2 - h^2 = H^2(1 - \hat{h})(1 + \hat{h})$:

$$F_{\text{gate}} = \frac{\rho g H^2 L}{2} (1 - \hat{h})(1 + \hat{h}) - \frac{\rho U^2 H L (1 - \hat{h})}{\hat{h}}.$$

From (c) the quadratic gives $F^2(1 + \hat{h}) = 2\hat{h}^2$:

$$F^2 = \frac{2\hat{h}^2}{1 + \hat{h}}.$$

Multiply by gH^2 :

$$U^2 H = \frac{2gH^2\hat{h}^2}{1 + \hat{h}}.$$

Substitute into the momentum term:

$$\frac{\rho U^2 H L (1 - \hat{h})}{\hat{h}} = \frac{2\rho g H^2 L \hat{h} (1 - \hat{h})}{1 + \hat{h}}.$$

Factor out $\rho g H^2 L (1 - \hat{h})/2$:

$$F_{\text{gate}} = \frac{\rho g H^2 L (1 - \hat{h})}{2} \left[(1 + \hat{h}) - \frac{4\hat{h}}{1 + \hat{h}} \right].$$

Combine over $(1 + \hat{h})$:

$$F_{\text{gate}} = \frac{\rho g H^2 L (1 - \hat{h})}{2} \cdot \frac{(1 + \hat{h})^2 - 4\hat{h}}{1 + \hat{h}}.$$

Expand the numerator:

$$(1 + \hat{h})^2 - 4\hat{h} = 1 + 2\hat{h} + \hat{h}^2 - 4\hat{h}.$$

Recognise $(1 - \hat{h})^2$:

$$(1 + \hat{h})^2 - 4\hat{h} = (1 - \hat{h})^2.$$

Substitute:

$$F_{\text{gate}} = \frac{\rho g H^2 L}{2} \frac{(1 - \hat{h})^3}{1 + \hat{h}}.$$

Checks. $\hat{h} \rightarrow 0$: recovers $\rho g H^2 L / 2$ as in (b). $\hat{h} \rightarrow 1$: force $\rightarrow 0$.

Question 2 — Potential flow and the complex potential $A\sqrt{z}$

(a)

Potential flow: velocity is the gradient of a scalar potential:

$$\mathbf{v} = \nabla \phi.$$

Vorticity vanishes identically:

$$\boldsymbol{\omega} = \nabla \times \mathbf{v} = \nabla \times \nabla \phi = 0.$$

Conditions to maintain irrotationality (Kelvin's theorem):

- (i) inviscid;
- (ii) barotropic, $\nabla p \parallel \nabla \rho$;
- (iii) conservative body forces;
- (iv) initially irrotational.

Incompressibility $\nabla \cdot \mathbf{v} = 0$ then yields Laplace's equation:

$$\nabla^2 \phi = 0.$$

(b)

Analyticity: dw/dz direction-independent. Approach along the real axis, $\Delta z = \Delta x$:

$$\frac{dw}{dz} = \frac{\partial \phi}{\partial x} + i \frac{\partial \psi}{\partial x}.$$

Approach along the imaginary axis, $\Delta z = i\Delta y$:

$$\frac{dw}{dz} = \frac{1}{i} \left(\frac{\partial \phi}{\partial y} + i \frac{\partial \psi}{\partial y} \right).$$

Use $1/i = -i$:

$$\frac{dw}{dz} = \frac{\partial \psi}{\partial y} - i \frac{\partial \phi}{\partial y}.$$

Equate real and imaginary parts:

$$\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}, \quad \frac{\partial \psi}{\partial x} = -\frac{\partial \phi}{\partial y}.$$

Cross-differentiating shows both are harmonic:

$$\nabla^2 \phi = \nabla^2 \psi = 0.$$

Use for 2D potential flow. Incompressibility $\partial_x v_x + \partial_y v_y = 0$ admits a streamfunction:

$$v_x = \frac{\partial \psi}{\partial y}, \quad v_y = -\frac{\partial \psi}{\partial x}.$$

Irrotationality $\partial_x v_y - \partial_y v_x = 0$ admits a potential:

$$v_x = \frac{\partial \phi}{\partial x}, \quad v_y = \frac{\partial \phi}{\partial y}.$$

Comparing gives the C-R relations: every 2D potential flow has analytic $w(z) = \phi + i\psi$. Complex velocity:

$$\frac{dw}{dz} = v_x - iv_y.$$

Sums, products and conformal maps of analytic functions remain analytic, building complex flow geometries from simple blocks.

(c)

Write $z = re^{i\theta}$ on the principal branch, $\theta \in [0, 2\pi)$:

$$w = A\sqrt{r}e^{i\theta/2}.$$

Split real/imaginary parts:

$$w = A\sqrt{r}\cos(\theta/2) + iA\sqrt{r}\sin(\theta/2).$$

Read off:

$$\phi = A\sqrt{r}\cos(\theta/2), \quad \boxed{\psi(r, \theta) = A\sqrt{r}\sin(\theta/2)}.$$

Streamlines $\psi = C$:

$$\sqrt{r}\sin(\theta/2) = C/A.$$

Square and set $a = C^2/A^2$:

$$r\sin^2(\theta/2) = a.$$

Apply $\sin^2(\theta/2) = (1 - \cos \theta)/2$:

$$r(1 - \cos \theta) = 2a.$$

Use $r \cos \theta = x$:

$$r - x = 2a.$$

Square using $r^2 = x^2 + y^2$:

$$(x + 2a)^2 = x^2 + y^2.$$

Expand and simplify:

$$y^2 = 4a(x + a).$$

Streamlines are confocal parabolas, focus at origin, sharing $\psi = 0$ on the positive x -axis.

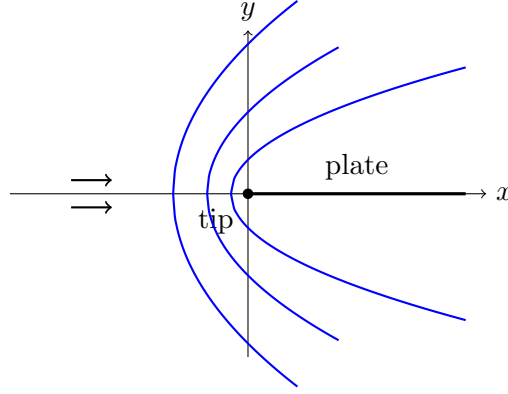
Physical flow. Flow around a semi-infinite plate (knife-edge) on the positive x -axis, fluid wraps around the tip at the origin. Complex velocity:

$$\frac{dw}{dz} = \frac{A}{2\sqrt{z}} = \frac{A}{2\sqrt{r}}e^{-i\theta/2}.$$

Speed diverges at the tip:

$$|\mathbf{v}| = \frac{A}{2\sqrt{r}}.$$

Streamline sketch.

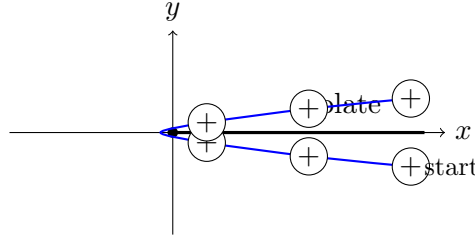


Each parabola enters from $\theta = 0$, wraps round the tip, returns along $\theta = 2\pi$.

Paddle wheel near $\theta = 2\pi^-$. Vorticity $\omega = 0$ in the bulk: the wheel *does not spin*, only advects. At $\theta \rightarrow 2\pi^-$, $e^{-i\theta/2} \rightarrow -1$:

$$v_x - iv_y = -\frac{A}{2\sqrt{r}}.$$

So $v_x < 0$, $v_y = 0$: wheel drifts along the underside towards the tip, whips around, and emerges along $\theta = 0^+$ with body axes fixed in the lab frame.



Orientation marker preserved: *no rotation in potential flow*.

(d)

Reynolds number:

$$\text{Re} = \frac{UL}{\nu}.$$

Compare Navier–Stokes terms:

$$\frac{\text{inertial}}{\text{viscous}} = \frac{U^2/L}{\nu U/L^2} = \text{Re}.$$

$\text{Re} \gg 1$: inertia dominates, viscosity confined to thin boundary layers. $\text{Re} \ll 1$: viscous (Stokes) flow.

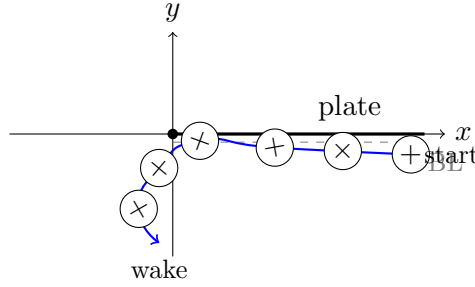
Two differences from high-Re real flow. (i) *Boundary layers*. Potential flow allows slip; real fluid demands $\mathbf{v} = 0$ at the wall. Viscosity sets a thin layer of thickness:

$$\delta \sim L \text{Re}^{-1/2} \ll L.$$

Wall-bounded vorticity $\sim U/\delta$ is continuously generated and convected downstream.

(ii) *Separation and wake*. Potential flow demands $|\mathbf{v}| \propto r^{-1/2} \rightarrow \infty$ at the tip. The real boundary layer cannot withstand the adverse pressure gradient: it *separates* at the tip, producing a turbulent wake of recirculating vortical fluid.

Real paddle wheel. Boundary layer vorticity $\omega \sim U/\delta \neq 0$ makes the wheel *spin* as it is convected. At the tip the boundary layer separates, shedding the wheel into the wake where it continues to tumble.



Orientation marker rotates progressively; wheel ends in the wake.

Why $\text{Re} \rightarrow \infty$ fails to recover potential flow. Viscosity $\nu \nabla^2 \mathbf{v}$ is a *singular perturbation*: it contains the highest spatial derivative. Dropping it lowers the PDE order, dropping the no-slip BC. With any $\nu > 0$, the boundary layer exists: $\delta \sim L \text{Re}^{-1/2} \rightarrow 0$, vorticity $\mathcal{O}(U/\delta) \rightarrow \infty$. Shed circulation per unit time stays $\mathcal{O}(1)$, producing wakes that persist in the limit (d'Alembert's paradox: finite drag at any Re).

Question 3 — Internal gravity waves

(a)

Assumptions. (i) Parcel keeps its own density $\rho_p = \rho_0(z_0)$ (incompressible, adiabatic). (ii) Pressure equilibrium (acoustic timescale $\ll$ buoyancy timescale). (iii) Small displacement ζ , Taylor-expand ρ_0 .

Newton's law per unit volume (buoyancy is net force):

$$\rho_p \ddot{\zeta} = -[\rho_p - \rho_0(z_0 + \zeta)]g.$$

Taylor-expand the background density:

$$\rho_0(z_0 + \zeta) \approx \rho_0(z_0) + \zeta \frac{d\rho_0}{dz}.$$

Use $\rho_p = \rho_0(z_0)$:

$$\rho_p - \rho_0(z_0 + \zeta) = -\zeta \frac{d\rho_0}{dz}.$$

Substitute into Newton's law:

$$\rho_0(z_0) \ddot{\zeta} = \zeta g \frac{d\rho_0}{dz}.$$

Divide by ρ_0 :

$$\ddot{\zeta} - \frac{g}{\rho_0} \frac{d\rho_0}{dz} \zeta = 0.$$

Define $N^2 = -(g/\rho_0) d\rho_0/dz$:

$$\ddot{\zeta} + N^2 \zeta = 0.$$

Read off the buoyancy frequency:

$$N = \sqrt{-\frac{g}{\rho_0} \frac{d\rho_0}{dz}}.$$

(b)

Decompose into background + perturbation:

$$\rho = \rho_0(z) + \delta\rho, \quad p = p_0(z) + \delta p.$$

Background hydrostatic balance:

$$\frac{dp_0}{dz} = -\rho_0 g.$$

Linearise Euler about rest, drop $\mathbf{v} \cdot \nabla \mathbf{v}$:

$$\rho_0 \frac{\partial \mathbf{v}}{\partial t} = -\nabla \delta p - \delta \rho g \hat{\mathbf{z}}.$$

Take the x -component:

$$\rho_0 \frac{\partial v_x}{\partial t} = -\frac{\partial \delta p}{\partial x}.$$

Take the z -component:

$$\rho_0 \frac{\partial v_z}{\partial t} = -\frac{\partial \delta p}{\partial z} - \delta \rho g.$$

Take ∂_z of the x -equation and ∂_x of the z -equation, subtract:

$$\frac{\partial}{\partial z} \left(\rho_0 \frac{\partial v_x}{\partial t} \right) - \frac{\partial}{\partial x} \left(\rho_0 \frac{\partial v_z}{\partial t} \right) = g \frac{\partial \delta \rho}{\partial x}.$$

Expand the LHS noting $\partial_x \rho_0 = 0$:

$$\rho_0 \frac{\partial}{\partial t} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) + \frac{d\rho_0}{dz} \frac{\partial v_x}{\partial t} = g \frac{\partial \delta \rho}{\partial x}.$$

Boussinesq: drop $d\rho_0/dz$ term (small by N^2/g):

$$\rho_0 \frac{\partial}{\partial t} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) = g \frac{\partial \delta \rho}{\partial x}.$$

Divide by ρ_0 and use $b = -g \delta \rho / \rho_0$:

$$\boxed{\frac{\partial}{\partial t} \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) = -\frac{\partial b}{\partial x}.$$

Interpretation. Horizontal buoyancy gradients tilt isopycnals and generate vorticity $\omega_y = \partial_z v_x - \partial_x v_z$ (baroclinic overturning).

(c)

Incompressibility $\partial_x v_x + \partial_z v_z = 0$ admits a streamfunction:

$$v_x = \frac{\partial \psi}{\partial z}, \quad v_z = -\frac{\partial \psi}{\partial x}.$$

Vorticity becomes:

$$\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} = \frac{\partial^2 \psi}{\partial z^2} + \frac{\partial^2 \psi}{\partial x^2} = \nabla^2 \psi.$$

Substitute into (b):

$$\frac{\partial}{\partial t} \nabla^2 \psi = -\frac{\partial b}{\partial x}.$$

Linearise $D\rho/Dt = 0$ about $\rho_0(z)$:

$$\frac{\partial \delta \rho}{\partial t} + v_z \frac{d\rho_0}{dz} = 0.$$

Multiply by $-g/\rho_0$:

$$\frac{\partial b}{\partial t} = \frac{g}{\rho_0} \frac{d\rho_0}{dz} v_z.$$

Use $N^2 = -(g/\rho_0) d\rho_0/dz$:

$$\frac{\partial b}{\partial t} = -N^2 v_z.$$

Substitute $v_z = -\partial_x \psi$:

$$\frac{\partial b}{\partial t} = N^2 \frac{\partial \psi}{\partial x}.$$

Differentiate the vorticity equation in t :

$$\frac{\partial^2}{\partial t^2} \nabla^2 \psi = -\frac{\partial}{\partial x} \frac{\partial b}{\partial t}.$$

Insert $\partial_t b = N^2 \partial_x \psi$:

$$\boxed{\frac{\partial^2}{\partial t^2} \left(\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial z^2} \right) = -N^2 \frac{\partial^2 \psi}{\partial x^2}.$$

(d)

Plane-wave ansatz $\psi \propto e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$:

$$\frac{\partial}{\partial x} \rightarrow ik_x, \quad \frac{\partial}{\partial z} \rightarrow ik_z, \quad \frac{\partial}{\partial t} \rightarrow -i\omega.$$

Substitute into the wave equation:

$$(-i\omega)^2 [(ik_x)^2 + (ik_z)^2] \psi = -N^2 (ik_x)^2 \psi.$$

Use $(-i\omega)^2 = -\omega^2$, $(ik_j)^2 = -k_j^2$, $k^2 = k_x^2 + k_z^2$:

$$\omega^2 k^2 = N^2 k_x^2.$$

Divide by k^2 , set $\cos \theta = k_x/k$:

$$\boxed{\omega^2 = N^2 \frac{k_x^2}{k_x^2 + k_z^2} = N^2 \cos^2 \theta.}$$

Frequency depends on direction only, bounded by N .

Group velocity. Take $\omega > 0$, $k_x > 0$, so $\omega = Nk_x/k$. Quotient rule:

$$\frac{\partial \omega}{\partial k_x} = \frac{N}{k} - \frac{Nk_x^2}{k^3}.$$

Simplify using $k^2 - k_x^2 = k_z^2$:

$$\frac{\partial \omega}{\partial k_x} = \frac{Nk_z^2}{k^3}.$$

Differentiate w.r.t. k_z :

$$\frac{\partial \omega}{\partial k_z} = -\frac{Nk_x k_z}{k^3}.$$

Assemble:

$$\mathbf{c}_g = \frac{Nk_z}{k^3} (k_z \hat{\mathbf{x}} - k_x \hat{\mathbf{z}}).$$

Dot with $\mathbf{k}$:

$$\mathbf{c}_g \cdot \mathbf{k} = \frac{Nk_z}{k^3} (k_x k_z - k_x k_z) = 0.$$

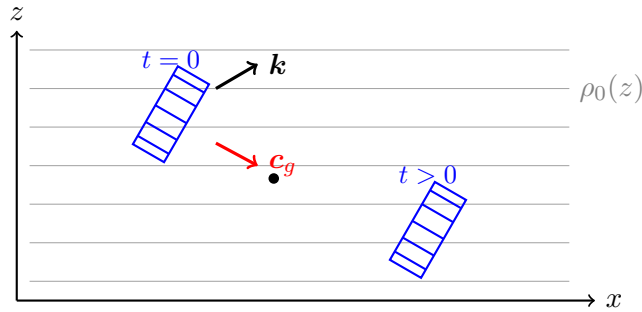
Hence:

$$\boxed{\mathbf{c}_g \perp \mathbf{k}},$$

i.e. $\mathbf{c}_g \perp \mathbf{c}_p$ since $\mathbf{c}_p \parallel \mathbf{k}$. Magnitude:

$$|\mathbf{c}_g| = \frac{N|k_z|}{k^2} = \frac{N|\sin \theta|}{k}.$$

Wave-packet sketch. For $k_x > 0$, $k_z > 0$ (figure), $\mathbf{c}_g \propto (k_z, -k_x)$ points down-and-right. Envelope translates along $\mathbf{c}_g$; phase fronts slide perpendicular to it (along $\mathbf{k}$).



Packet retains its tilt (all $\mathbf{k}$ share the same θ , hence same $\mathbf{c}_g$). Phase fronts slide through the envelope.

Question 4 — Thin-film viscous flow on an inclined plane

(a)

Axes: x down-slope, z normal; plate $z = 0$, surface $z = h(x, t)$. Gravity:

$$\mathbf{g} = g \sin \theta \hat{\mathbf{x}} - g \cos \theta \hat{\mathbf{z}}.$$

Lubrication assumptions. (i) Thin film $\varepsilon \equiv h/L \ll 1$. (ii) Reduced Reynolds $\varepsilon \text{Re} \ll 1$ (inertia negligible). (iii) Incompressible.

Compare viscous second derivatives:

$$\frac{\partial^2 u}{\partial x^2} \sim \frac{U}{L^2}, \quad \frac{\partial^2 u}{\partial z^2} \sim \frac{U}{h^2}.$$

Ratio $h^2/L^2 = \varepsilon^2$: keep $\partial_z^2 u$ only. Steady NS along the plate:

$$0 = -\frac{1}{\rho} \frac{\partial p}{\partial x} + g \sin \theta + \nu \frac{\partial^2 u}{\partial z^2}.$$

Normal to the plate (all dynamical terms negligible):

$$0 = -\frac{1}{\rho} \frac{\partial p}{\partial z} - g \cos \theta.$$

Boundary conditions.

- No-slip: $u(x, 0, t) = 0$, $w(x, 0, t) = 0$.
- No tangential stress at free surface: $\partial_z u|_{z=h} = 0$.
- Atmospheric pressure at free surface: $p(x, h, t) = p_{\text{atm}}$.
- Kinematic condition: $\partial_t h + u \partial_x h = w$ at $z = h$.

(b)

Integrate $\partial_z p = -\rho g \cos \theta$ from z to h with $p(h) = p_{\text{atm}}$:

$$p(x, z, t) = p_{\text{atm}} + \rho g \cos \theta [h(x, t) - z].$$

Differentiate in x :

$$\frac{\partial p}{\partial x} = \rho g \cos \theta \frac{\partial h}{\partial x}.$$

Substitute into x -momentum:

$$\nu \frac{\partial^2 u}{\partial z^2} = \frac{1}{\rho} \frac{\partial p}{\partial x} - g \sin \theta.$$

Simplify:

$$\nu \frac{\partial^2 u}{\partial z^2} = -g \left(\sin \theta - \cos \theta \frac{\partial h}{\partial x} \right).$$

Integrate once in z :

$$\nu \frac{\partial u}{\partial z} = -g \left(\sin \theta - \cos \theta \frac{\partial h}{\partial x} \right) z + C_1.$$

Apply $\partial_z u|_{z=h} = 0$:

$$C_1 = gh \left(\sin \theta - \cos \theta \frac{\partial h}{\partial x} \right).$$

Substitute back:

$$\nu \frac{\partial u}{\partial z} = g \left(\sin \theta - \cos \theta \frac{\partial h}{\partial x} \right) (h - z).$$

Integrate again, $C_2 = 0$ from $u(0) = 0$:

$$u(x, z, t) = \frac{g}{\nu} \left(\sin \theta - \cos \theta \frac{\partial h}{\partial x} \right) \left(hz - \frac{1}{2} z^2 \right).$$

Volumetric flux per unit width.

$$q(x, t) = \int_0^h u \, dz.$$

Pull out prefactor:

$$q = \frac{g}{\nu} \left(\sin \theta - \cos \theta \frac{\partial h}{\partial x} \right) \int_0^h \left(hz - \frac{1}{2} z^2 \right) dz.$$

Evaluate the integral:

$$\int_0^h \left(hz - \frac{1}{2} z^2 \right) dz = \frac{1}{2} h^3 - \frac{1}{6} h^3 = \frac{h^3}{3}.$$

Hence:

$$q = \frac{gh^3}{3\nu} \left(\sin \theta - \cos \theta \frac{\partial h}{\partial x} \right).$$

Mass conservation. Integrate $\partial_x u + \partial_z w = 0$ from 0 to h , $w(0) = 0$:

$$w(h) = - \int_0^h \frac{\partial u}{\partial x} dz.$$

Apply Leibniz on q :

$$\frac{\partial q}{\partial x} = \int_0^h \frac{\partial u}{\partial x} dz + u(x, h, t) \frac{\partial h}{\partial x}.$$

Combine:

$$w(h) = - \frac{\partial q}{\partial x} + u(x, h, t) \frac{\partial h}{\partial x}.$$

Apply the kinematic BC $\partial_t h + u(h) \partial_x h = w(h)$:

$$\frac{\partial h}{\partial t} = - \frac{\partial q}{\partial x}.$$

Substitute q :

$$\boxed{\frac{\partial h}{\partial t} + \frac{\partial}{\partial x} \left[\frac{gh^3}{3\nu} \left(\sin \theta - \cos \theta \frac{\partial h}{\partial x} \right) \right] = 0.}$$

(c)

For $\theta = \mathcal{O}(1)$ with $\cos \theta |\partial_x h| \ll \sin \theta$, drop the diffusive correction:

$$\frac{\partial h}{\partial t} + \frac{\partial}{\partial x} \left(\frac{gh^3 \sin \theta}{3\nu} \right) = 0.$$

Apply the chain rule:

$$\frac{\partial h}{\partial t} + \frac{g \sin \theta}{3\nu} 3h^2 \frac{\partial h}{\partial x} = 0.$$

Recast as quasilinear:

$$\frac{\partial h}{\partial t} + u(h, \theta) \frac{\partial h}{\partial x} = 0,$$

with the kinematic-wave speed

$$\boxed{u(h, \theta) = \frac{gh^2 \sin \theta}{\nu}.}$$

This equals the surface velocity $u(z = h)$ from (b), three times the depth-averaged $\bar{u} = q/h = gh^2 \sin \theta / (3\nu)$.

Solve by characteristics. On $dx/dt = u(h, \theta)$:

$$\frac{dh}{dt} = \frac{\partial h}{\partial t} + u \frac{\partial h}{\partial x} = 0.$$

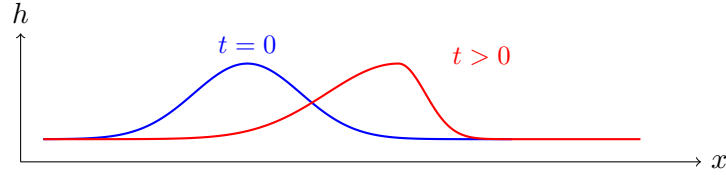
So h is constant along straight characteristics:

$$x = x_0 + u(h_0, \theta) t.$$

Eliminate x_0 :

$$h(x, t) \approx h_0 \left(x - \frac{gh^2 \sin \theta}{\nu} t \right).$$

Bump evolution. Wave speed $u(h, \theta) \propto h^2$ increases with h . Thicker fluid catches up with thinner fluid ahead, so the forward face *steepens*; the rear face *flattens*:



Characteristics from the leading slope cross at a finite time t^* : $\partial_x h$ becomes singular (shock). At that moment the kinematic-wave approximation breaks down, since the neglected $\cos \theta \partial_x h$ term is no longer small and regularises the shock into a finite-width front.